
Automotive SoC Motor Driver

1.0 GENERAL PURPOSE INPUT OUTPUT (GPIO)

1.1 Features

- The A89201 has a total of 16 general purpose I/O pins: PD0 to PD15
- Out of the 16 GPIO pins, 8 of them are bidirectional and available for external connection. The other 8 pins are used for internal die to die connection and are unidirectional.
- Configurable pin function
- Configurable passive pull-up/pull-down settings
- 8 configurable modes of operation
- 16 selectable external output driver sources for each pin
- 8 unique selectable external sources for each pin for debugging purposes and die-to-die interfaces
- Configurable output driver
- Configurable drive strength
- Short-to-GND and short-to-supply protection
- Data transfer through digital inputs or outputs
- Analog voltage measurement by configuring the desired pin as analog input.
- Perform input/output functions for on-chip peripherals
- Configurable interrupt sources with interrupt enables for input signals

1.2 Introduction

The A89201 features 16 GPIO pins on board. GPIO pins, PD7 to PD0, are bidirectional and used for external connection. The GPIO pins, PD15 to PD8, are uni-directional and used for internal die to die connection. A correct pin configuration for PD15 to PD8 is required to ensure correct die to die operation. More information about PD15 – PD8 configuration settings can be found in [section 1.5](#). PD7 to PD0 can be configured as input or output independently. Each pin is also connected to several internal (pull-up, pull-down, push-pull, pull-down open drain...etc) circuits that can be selected individually. The module also supports configurable interrupt sources with individual enable/disable capability for each pin. Short circuit protection is also featured when the pin is configured as an output to prevent any damages due to a short circuit. The blanking time is available when the pin is configured as output to avoid potential short-circuit missdetection. Interrupt generation is also possible in case of detecting short-circuit when the corresponding setting is enabled.

1.3 Functional Description

The general block diagram of the GPIO module is shown in Figure 1-1. There are a total of 16 GPIOs, PD15 to PD0, out of which 8 are used for external connection, and 8 of them are used for die-to-die interface. PD7 to PD0 can be configured as either input or output to receive an external signal or generate a desired source. PD15 to PD10 are used to define the logic level signals for the die-to-die gate driver interface and must be configured as output. PD9 and PD8 are used for the die-to-die LIN interface. PD9 must be configured as output as it defines the LIN-Tx signals to the LIN-transceiver and PD8 must be configured as input as it defines LIN-Rx signal received from the LIN-transceiver. More information about configuring PD15-PD8 are provided in [section 1.5](#).

If the pin is configured as an output, a short-circuit protection system is implemented on the driven output; If a short circuit is detected, the pin will automatically flip to input mode and an IRQ is generated to inform the host of the detected fault. The source used to generate the output is configurable and explained in details in the following sections.

If the pin is configured as an input, the GPIO module will synchronize the external input signal to the internal logic. The interrupt system can be configured to generate an interrupt request on a detection of edge-selection event.

The interrupt system generates a single active-high interrupt request signal to the Cortex M4's NVIC. The interrupt source may be a fault detection (short circuit) on the output GPO or an edge detection event on the input signal GPI. All the interrupt sources are configurable and maskable.

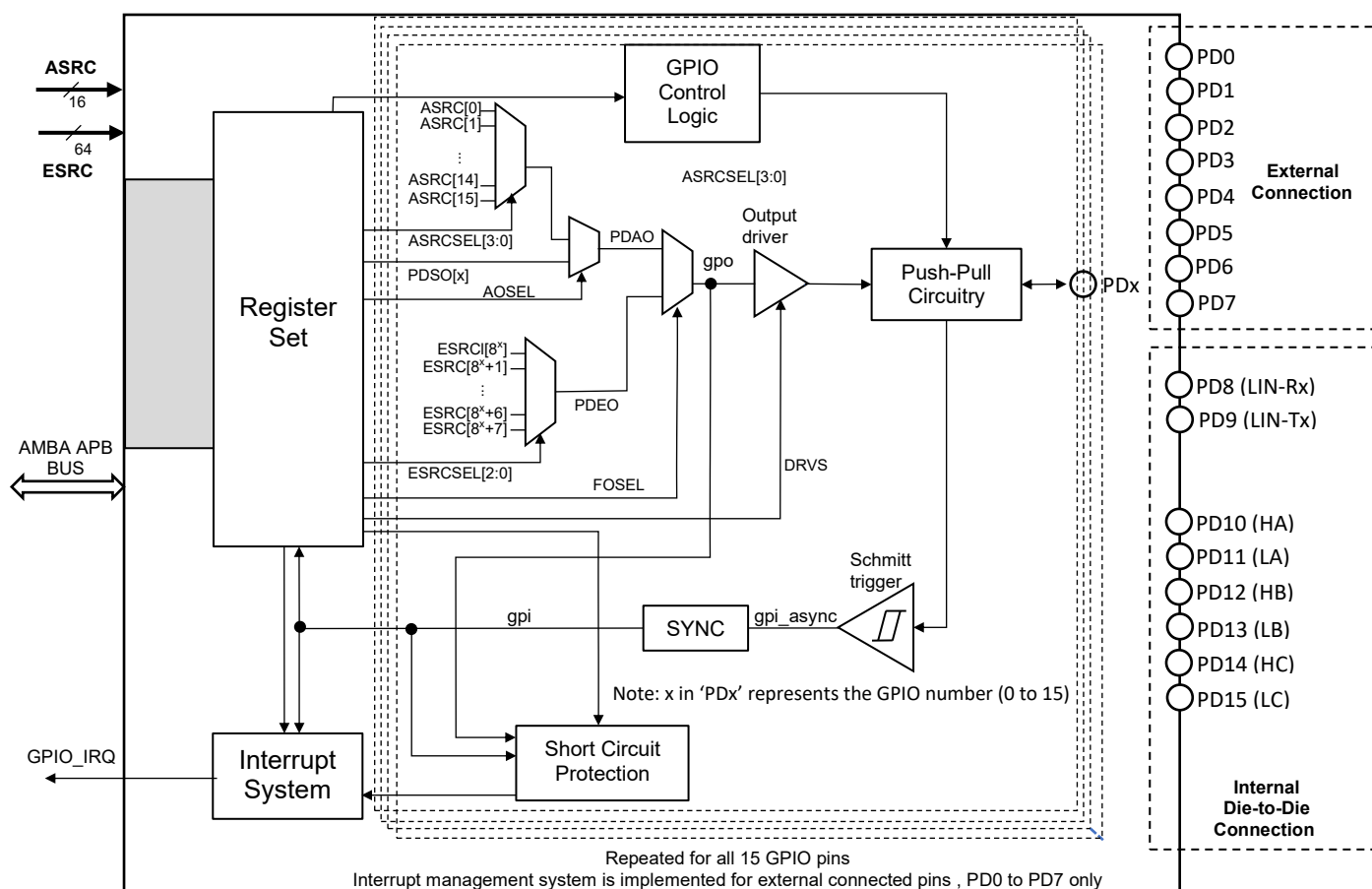


Figure 1-1: General block diagram of the GPIO module

1.4 Electrical Characteristics

Table 1: EC Table: Valid at $T_J = \text{TBD}$, $4.0\text{V} \leq \text{VD50} \leq 5.5\text{V}$, $1.08\text{V} \leq \text{VDD} \leq 1.32\text{V}$

Characteristics	Symbol	Test Conditions	Min.	Typ.	Max.	Unit
Input low voltage	V_{IL}		-	-	$0.3 \times V_{IO}$	V
Input high voltage	V_{IH}		$0.7 \times V_{IO}$	-	-	V
Input hysteresis	V_{HYS}		200	-	-	mV
Input pull-up resistor	R_{PU}		30	50	70	$k\Omega$
Input pull-down resistor	R_{PD}		30	50	70	$k\Omega$
Input leakage	I_{ILK}	$0\text{ V} < V_{IN} < V_{IO}$	-1	-	1	μA
Output low voltage	V_{OL}	$I_{OL} = -1\text{ mA}$	0	-	400	mV
		$I_{OL} = -4\text{ mA}$				
Output high voltage	V_{OH}	$I_{OH} = 1\text{ mA}$	-400	-	0	mV
		$I_{OH} = 4\text{ mA}$				
Output leakage	I_{OLK}	$0\text{ V} < V_O < V_{IO}$	-1	-	1	μA

- If 5V operation is desired on a GPIO pin, VIO pin needs to be connected to VD50, otherwise GPIO pin operates at 3.3V.

1.5 Ports IN/OUT Settings

1.5.1 Input Selection

The GPIO pins can be configured as an input when PIN_MODE in GPIO_CONTROL register is set to one of the input modes. The available input modes are digital input pull-up, digital input pull-down, general digital input, and analog input. When configured as digital input, the logic level of the signal connected to GPIO pin can be read back by GPIL paramter GPIO_PDIN register. It is also possible to generate an interrupt on the rising,falling, rising and falling edge event, high level or low level of the connected external signals. The detection event can be configured by GPI_INTCFG paramter and the interrupt can be enabled by configuring GPI_INTEN bit in GPIO_CONTROL register. The pins PD7 to PD0 are also connected to data acquisition unit (DAU) peripheral to measure the analog voltages. Therefore if it is desired to measure the analog voltages connected to these pins, PIN_MODE must be set to analog input.

Important Notes:

- (1) PD8 is used for LIN-Rx signal in the die-to-die interface and must always be configured as digital input (PIN_MODE = 0001_b).
- (2) PD9 to PD15 are used as output for die-to-die interface and **must never** be configured as input. The correct settings for these pins are proposed in the following section.

1.5.2 Output Selection

The GPIO pins can be configured as output when PIN_MODE in GPIO_CONTROL register is set to one of the output modes. The available output modes are digital output push-pull, push-down open drain, and push-up open drain. Once a pin is configured as an output, the output can be either forced from software, or it can be generated from the external sources generated by other peripherals such as PWMU. Among the sources, the ones that are common in many applications, are shared and similar across all the GPIO pins and are defined by ASRC array. A corresponding source from ASRC array can be selected by configuring ASRCSEL paramter in the GPIO_CONTROL register. On the other hand, there are some sources which are used for debugging purposes only or must be tied to a specific pin in order to operate correctly (ie: die to die connection sources). These sources are sepcific for each GPIO pin and are not shared across all the GPIOs and are defined by ESRC array. A corresponding source from ESRC array can be selected by configuring ESRCSEL parameter in GPIO_USRC register. ESRC array contains a total of 127 sources and ASRC array contains a total of 15 sources. Table below summerize how to choose a desired source for the pin output which are decided by AOESL and FOSEL paramters in GPIO_CONTROL and GPIO_USRC registers respectively.

Table 2: General Purpose Output Truth Table Logic

AOSEL	FOSEL	GPO
x	1	Select from ESRC Array (configured by ESRCSEL)
0	0	PDSO (controlled by software by programming GPIO_PDOUT register)
1	0	Select from ASRC array(configured by ASRCSEL)

Important Note:

- (1) PD9 is used for LIN-Tx die-to-die interface. This pin must always be configured as digital output (PIN_MODE = '0101_b'), and the source must be generated by ASRC (AOSEL = 1 & FOSEL = 0). In this case ASRCSEL must be set to '1000_b' which corresponds to the LIN-Tx signal.

- (2) PD10 to PD15 are used for the gate driver logic signals die-to-die interface and must always be configured as output (PIN_MODE = '0101_b'),. The gate driver logic level signals are generated by pulse width modulation unit (PWMU) and placed in ESRC array. Therefore, ESRC array must be used for these pins (FOSEL =1). ESRCSEL must be set to '0000_b' to ensure a correct connection of signals from pulse width modulation unit (PWMU) to the gate driver.

The figures below show the analog circuitry of GPIO pins in each of the operation modes:

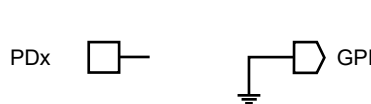


Figure 1-2: GPIO High-Z mode (PIN_MODE = 0x0)

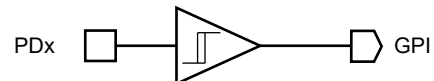


Figure 1-3: GPIO digital input mode (PIN_MODE = 0x1)

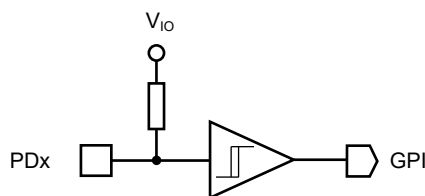


Figure 1-4: GPIO digital input pull-up mode (PIN_MODE = 0x2)

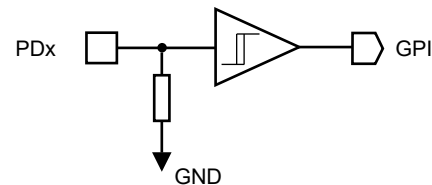


Figure 1-5: GPIO digital input pull-down mode (PIN_MODE = 0x3)

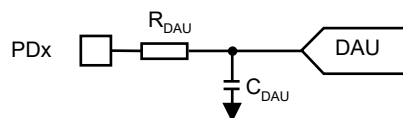


Figure 1-6: GPIO analog input mode (PIN_MODE = 0x4)

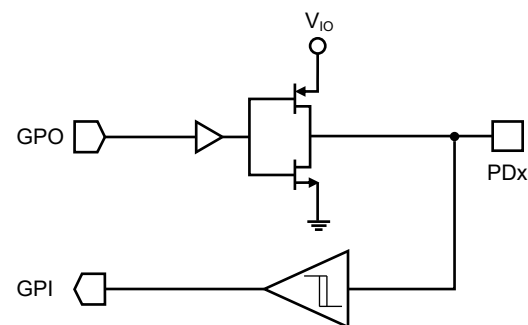


Figure 1-7: GPIO output push-pull mode (PIN_MODE = 0x5)

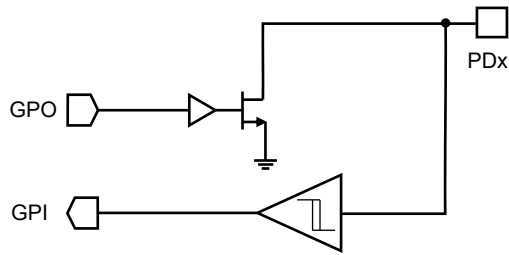


Figure 1-8: GPIO output pull-down open-drain mode (PIN_MODE = 0x6)

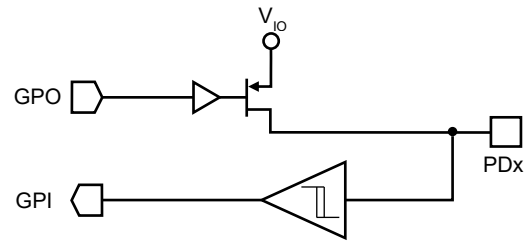


Figure 1-9: GPIO output pull-up open-drain mode (PIN_MODE = 0x7)

1.6 Short Circuit Protection

The GPIO module features a short-circuit protection scheme for external GPIO pins, PD0 to PD7 when configured as an output. The module will detect if a short circuit occurs either to supply or GND and will automatically flip the pin's operation mode to digital input mode (PIN_MODE = 0x1). The pin is held in this operation mode until the fault is cleared.

The fault detection capabilities the GPIO_BLANKx registers. Depending on the configuration in these registers, the fault detection logic waits for the programmed blanking-time window and then checks the levels on both GPO and GPI lines. Ideally, when the pin is operating as an output and is being driven to a certain logic level, the same level should be read on the GPI line. If there is a mismatch between those two lines, then a short-circuit is detected. Depending on the level of GPI, the fault source is distinguished: If GPI is stuck at high logic level, when GPO is commanded to a low logic level, a short-to-supply fault is flagged. If the GPI is stuck at low logic level, when GPO is commanded to a high logic level, a short-to-GND fault is flagged.

When a fault is detected, a flag is set to report the fault to the host controller. A short-to-GND fault on a given pin will set the corresponding bit in parameter STG in the GPIO_STG register. Similarly, a short-to-supply fault on a given pin will set the corresponding bit in parameter STS in the GPIO_STS register. If the interrupt generation option is enabled in GPIO_STS_INTEN or GPIO_STG_INTEN registers, once a STS or STG fault flag is set, an interrupt request (IRQ) is set to interrupt the host controller.

If a fault is detected, the faulty pin will be automatically switched to digital input mode. The pin remains locked in this mode until the fault is cleared in the GPIO_STG or GPIO_STS registers. To clear the fault flag, the user software shall write '1' to the asserted bit after removing the short condition. If the flags are reset but the fault condition is not removed, the short-circuit protection logic will reset the blanking-time window and wait again for that duration and regenerates the flag if the fault condition is still present.

To switch off the short-circuit protection scheme, the blanking time window of the desired pin in the GPIO_BLANKx registers shall be set to 0.

Figure 1-10 shows an example of a short-to-GND fault detection.

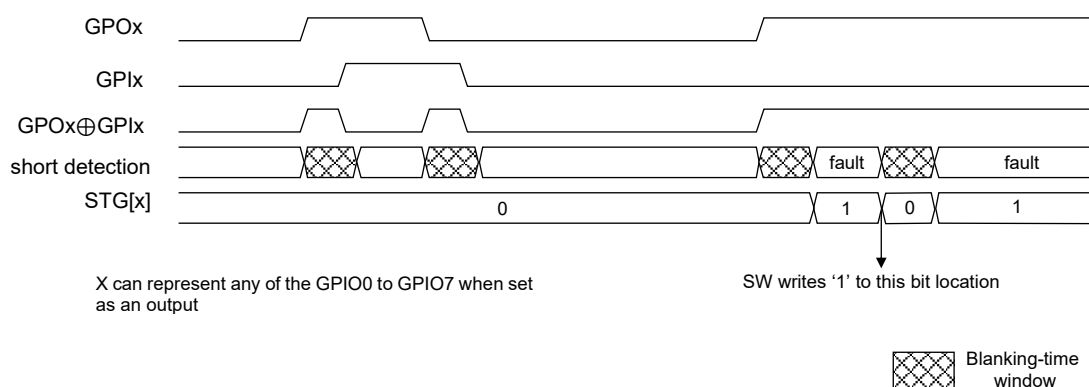


Figure 1-10: GPIO short-to-GND fault detection

Figure 1-11 shows an example of a short-to-supply fault detection.

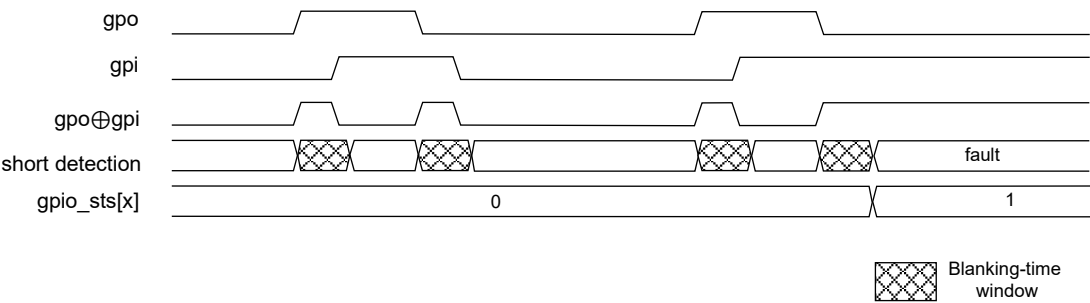


Figure 1-11: GPIO short-to-supply fault detection

1.7 Interrupt Management

The GPIO generates a single active-high interrupt request signal (IRQ) to the Cortex M4's NVIC. Depending on the mode of operation, the interrupt source may vary.

If the GPIO pin is configured as a digital input, an interrupt request can be generated based on any event detected on the input signal GPI. The desired event is selected by the GPI_INTCFG bits in the pin's corresponding GPIO_CONTROLx register. An interrupt can be generated on detecting a rising edge, falling edge, rising or falling edge, level high or a level low on the input signal.

In order to generate an interrupt request, the option must be enabled via the GPI_INTEN bit in the corresponding GPIO_CONTROLx register. Upon detecting the selected event, the corresponding bit in EDS paramter in GPIO_EVENT_STATUS register is set, indicating that the event selected by the GPI_INTCFG bits has been detected. Each bit in EDS corresponds to a GPIO pin, PDx. Bit 0, EDS[0] corresponds to PD0, bit 1, EDS[1], corresponds to PD1, etc.

If a GPIO pin is configured as a digital output, the interrupt generation on the GPI signal is locked by hardware and GPI_INTCFG and GPI_INTEN bits in the GPIO_CONTROLx register become obsolete. When configured as an output, if a short circuit condition is detected on any of the PD0 to PD7 pins, it is reported in GPIO_STG or GPIO_STS registers. It is also possible to generate interrupt when a short circuit occurs, however, the interrupt generation feature must be enabled via the GPIO_STG_INTEN and GPIO_STS_INTEN registers.

Interrupt requests are cleared by writing '1' to the corresponding bit in GPIO_EVENT_STATUS, GPIO_STS or GPIO_STG registers. Figure 1-12 and Figure 1-13 show the interrupt generation logic of the GPIO module.

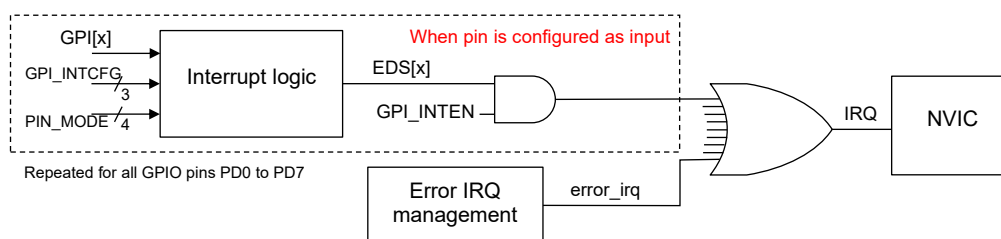


Figure 1-12: Interrupt generation logic of a GPIO pin

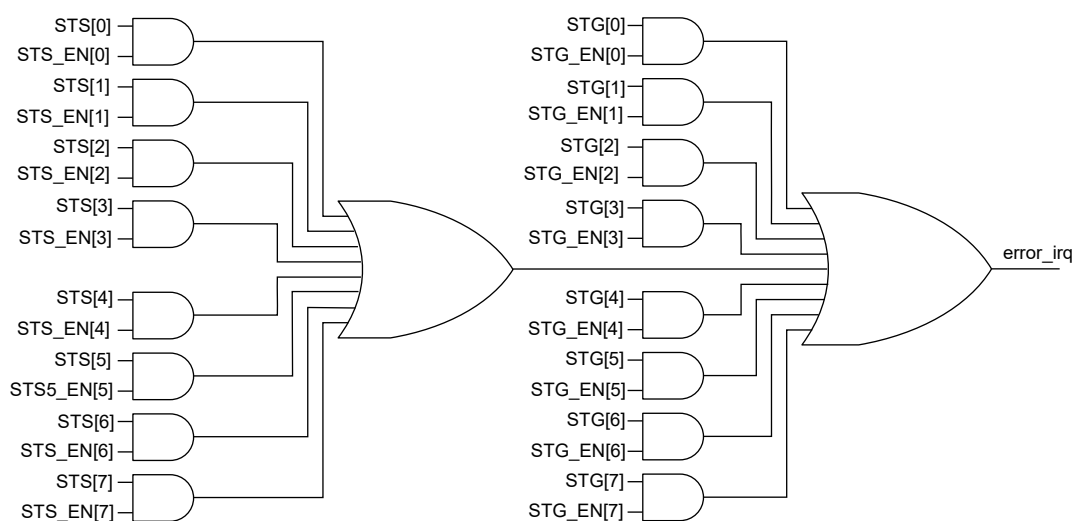


Figure 1-13: Error interrupt request generation logic

1.7.1 Examples on interrupt generation

This section shows some examples of interrupt generation when the pin is configured to operate as either an input or an output.

1.7.1.1 Digital input: interrupt on a rising edge detection

In this example, shown in Figure 1-14, GPIO pin 0 operates as a digital input. The pin is configured to generate an interrupt request on the detection of a rising edge on the input signal connected to PD0. This is done by configuring the GPI_INTCFG bits in the GPIO_CONTROL0 register and enabling the interrupt generation feature via the GPI_INTEN bit in the same register.

When a rising edge is detected on the input signal, EDS[0] in GPIO_EVENT_STATUS register gets set to 1, indicating that the configured event has occurred. The IRQ will be generated if GPI_INTEN is set to 1 as shown in Figure 1-12. The GPIO_IRQ is reset by writing '1' to its bit location in the gpio_intstat register via software. When this bit is reset, the IRQ signal goes low as well. It is crucial that the software resets this bit in time as to not miss subsequent events on the input signal.

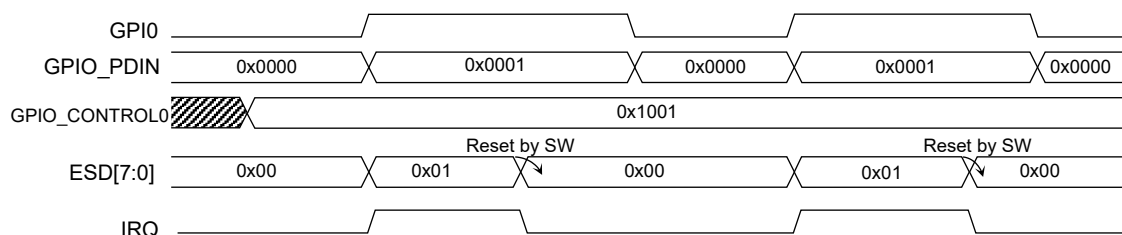


Figure 1-14: Interrupt on a rising edge detection

1.7.1.2 Digital input: interrupt on a level-high detection

In this example, shown in Figure 1-15, GPIO pin 0, PD0, is configured as a digital input. An interrupt request is generated on the detected of a level-high on the input GPIO. The module behaves similarly as described in section 1.7.1.1. However, as the interrupt is generated by a level-high detection, the hardware will set the status bit ESD[0] in the GPIO_EVENT_STATUS register once again after the software clears it, if the condition remains true after the software 'write 1 to clear' to GPIO_EVENT_STATUS register is finished.

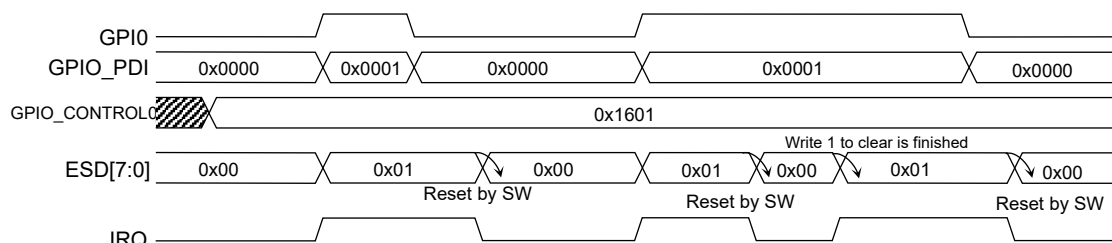


Figure 1-15: Interrupt generation on the detection of a level-high input

1.7.1.3 Digital input: Multiple interrupt requests

In this example, shown in Figure 1-16, GPIO pins PD0 and PD1 are configured as digital inputs. Both pins are configured to generate an interrupt request on the detection of a rising edge on their respective input signals GPIO and GPI1.

When an event is detected on one of the pins, the corresponding bit in the GPIO_EVENT_STATUS register is set. The IRQ signal goes high following any of the asserted bits in this register as explained in Figure 1-12. If the SW does not clear the asserted bit in GPIO_EVENT_STATUS register before the detection of an interrupt event on another pin, the interrupt flags GPIO_IRQ and GPI1_IRQ will stack as both will be set. The IRQ will remain high as it is a logical OR gate of both bits. To clear the interrupt, both bits ESD[0] and ESD[1] in GPIO_EVENT_STATUS register must be cleared by writing '1' to each bit's location.

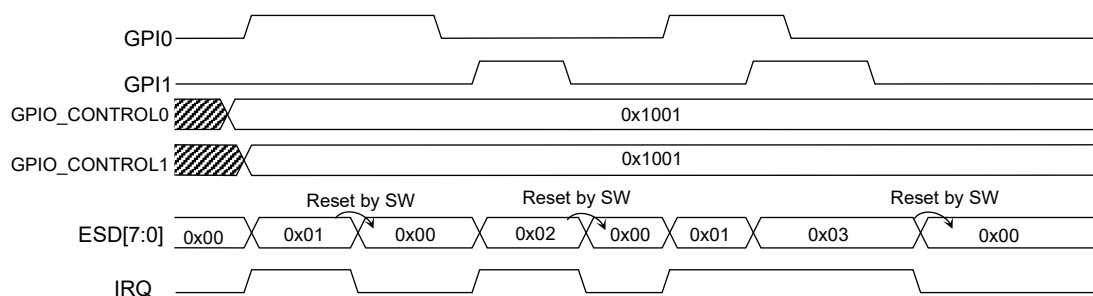


Figure 1-16: Multiple interrupt requests

1.7.1.4 Digital output: Short-to-GND fault detection

In this example, shown in Figure 1-17, GPIO pin 0, PDO, is configured to operate as a digital output. Thus, the short-circuit protection logic scans for any faults detected on the pin.

The short-circuit detection feature is enabled by setting the corresponding blanking-time bits specified in the GPIO_BLANK0 to a non-zero value.

When the blanking time-window expires, the logic checks for any discrepancies between the level on GPIO and GPO0. As shown in the Figure 1-17, once the fault is detected, the corresponding bit, STG[0], is set in the GPIO_STG register. The IRQ signal is set high following the asserted bit in GPIO_STG register as described in Figure 1-13. The IRQ generation is enabled by setting STG_EN[0] bit in the GPIO_STG_INTEN register.

If the user SW clears the fault flag in the GPIO_STG register by writing 1 to the asserted bit, the blanking window will be reset and after it expires a second time, if the fault condition is still present on the pin, the short-detection logic will set the corresponding STG[x] flag once again, which, in turn, will set the IRQ signal high. The user must ensure that the short condition is removed before clearing the fault flag.

The short-to-supply fault detection follows the same procedure described in this example.

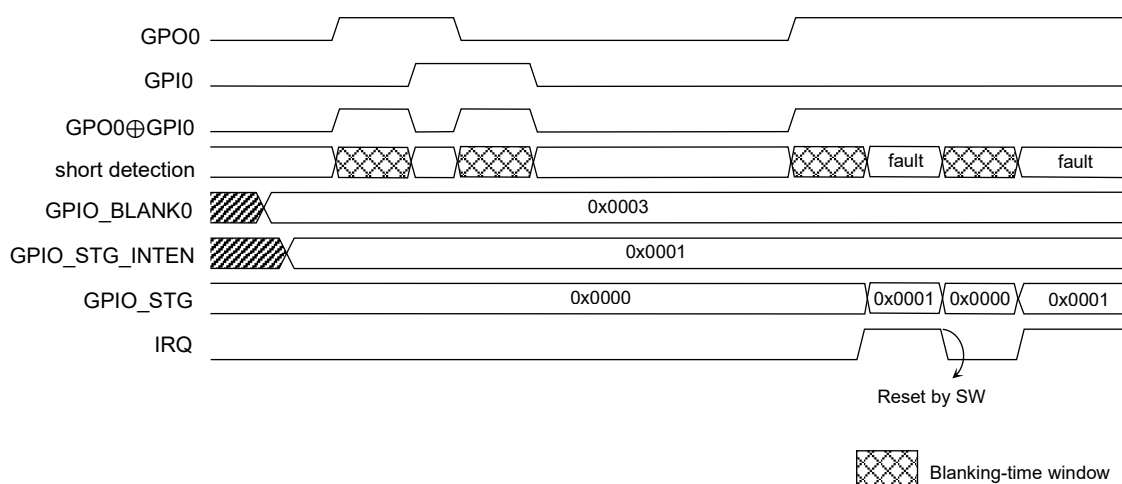


Figure 1-17: Interrupt generation on a short-to-GND fault detection

1.8 Register Set

Each GPIO pin has its own control (GPIO_CONTROLx) and unique source selection (GPIO_USRCx) register to configure its behavior. In addition, blanking registers are used to configure the short-circuit protection functionality for GPIO pins PD0 to PD7. Input and output data are stored in the generic input and output registers respectively. Interrupt management registers enable the user to enable or disable interrupt sources as well as check the state of the module.

The register map of the GPIO consists of 31 16-bit registers. Registers are word aligned. unimplemented bits are read as 0 and reserved bits can have undeterministic value. Interrupts are cleared by writing 1 to the corresponding status bit. Table 3 summarizes the register addresses, access types and reset values. The starting address of the GPIO module is 0x40008000.

Base Address: 0x40008000

Table 3: Register map of the GPIO module

Register Name	Access Type	Address Offset	Reset Value
GPIO_CONTROL0	R/W	0x00	0x0000
GPIO_CONTROL1	R/W	0x04	0x0000
GPIO_CONTROL2	R/W	0x08	0x0000
GPIO_CONTROL3	R/W	0x0C	0x0000
GPIO_CONTROL4	R/W	0x10	0x0000
GPIO_CONTROL5	R/W	0x14	0x0000
GPIO_CONTROL6	R/W	0x18	0x0000
GPIO_CONTROL7	R/W	0x1C	0x0000
GPIO_CONTROL8	R/W	0x20	0x0000
GPIO_CONTROL9	R/W	0x24	0x0000
GPIO_CONTROL10	R/W	0x28	0x0000
GPIO_CONTROL11	R/W	0x2C	0x0000
GPIO_CONTROL12	R/W	0x30	0x0000
GPIO_CONTROL13	R/W	0x34	0x0000
GPIO_CONTROL14	R/W	0x38	0x0000
GPIO_CONTROL15	R/W	0x3C	0x0000
GPIO_BLANK0	R/W	0x40	0x0000
GPIO_BLANK1	R/W	0x44	0x0000
Reserved	R/W	0x48	0x0000
Reserved	R/W	0x4C	0x0000
GPIO_PDIN	R	0x50	0x0000
GPIO_PDOUT	R/W	0x54	0x0000
GPIO_STG	R/1C	0x58	0x0000
GPIO_STS	R/1C	0x5C	0x0000
GPIO_STG_INTEN	R/W	0x60	0x0000
GPIO_STS_INTEN	R/W	0x64	0x0000
GPIO_EVENT_STATUS	R/1C	0x68	0x0000
GPIO_USRC0	R/W	0xFC0	0x0000
GPIO_USRC1	R/W	0xFC4	0x0000
GPIO_USRC2	R/W	0xFC8	0x0000
GPIO_USRC3	R/W	0xFCC	0x0000
GPIO_USRC4	R/W	0xFD0	0x0000
GPIO_USRC5	R/W	0xFD4	0x0000
GPIO_USRC6	R/W	0xFD8	0x0000
GPIO_USRC7	R/W	0xFDC	0x0000
GPIO_USRC8	R/W	0xFE0	0x0000
GPIO_USRC9	R/W	0xFE4	0x0000
GPIO_USRC10	R/W	0xFE8	0x0000
GPIO_USRC11	R/W	0xFEC	0x0000
GPIO_USRC12	R/W	0xFF0	0x0000
GPIO_USRC13	R/W	0xFF4	0x0000
GPIO_USRC14	R/W	0xFF8	0x0000
GPIO_USRC15	R/W	0xFFC	0x0000

GPIO_CONTROLx

Reset value = 0x0000

x represent the GPIO pin number.

ie: GPIO_CONTROL0 setting is for PD0

Register Address GPIO_CONTROL0 = 0x40008000
 Register Address GPIO_CONTROL1 = 0x40008004
 Register Address GPIO_CONTROL2 = 0x40008008
 Register Address GPIO_CONTROL3 = 0x4000800C
 Register Address GPIO_CONTROL4 = 0x40008010
 Register Address GPIO_CONTROL5 = 0x40008014
 Register Address GPIO_CONTROL6 = 0x40008018
 Register Address GPIO_CONTROL7 = 0x4000801C
 Register Address GPIO_CONTROL8 = 0x40008020
 Register Address GPIO_CONTROL9 = 0x40008024
 Register Address GPIO_CONTROL10 = 0x40008028
 Register Address GPIO_CONTROL11 = 0x4000802C
 Register Address GPIO_CONTROL12 = 0x40008030
 Register Address GPIO_CONTROL13 = 0x40008034
 Register Address GPIO_CONTROL14 = 0x40008038
 Register Address GPIO_CONTROL15 = 0x4000803C

This register defines if a GPIO pin should be used as input or output. It also configures what should be the output source on a GPIO pin, when FOSEL in GPIO_USRC register is set to 0. In addition it configures the trigger event condition that GPIO should detect when set as an input and if an interrupt should be generated upon receiving the trigger event condition. The gate drive strength is also configured by this register. GPIO_CONTROL0 to GPIO_CONTROL7 can be configured with any desired setting. However, GPIO_CONTROL8 to GPIO_CONTROL15 must be configured with a fixed setting as they are used for die-to-die interface. For correct configuration setting for GPIO8 to GPIO15 read input/output selection [section](#).

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	AOSEL	GPI_INTEN	GPI_INTCFG[2:0]		DRVS	
U	U	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
ASRCSEL[3:0]				PIN_MODE[3:0]			
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
PIN_MODE[3:0]	3:0	Select the GPIO mode of the pin PDx¹. 0000: High-Z (default) 0001: Digital input 0010: Digital input pull-up 0011: Digital input pull-down 0100: Analog input (Input to DAU module) 0101: Digital output push-pull 0110: Digital output pull-down open-drain 0111: Digital output pull-up open-drain 1xxx: High-Z

Note: (1) x represent the GPIO pin number ranging from 0 to 15.

ASRCSEL[3:0]	7:4	Application Source (ASRC) Selection ¹. 0000b: SCIO_Tx (default)² 0001b: SCI1_Tx⁽²⁾ 0010b: SPI0_SCKOUT⁽²⁾ 0011b: SPI1_SCKOUT⁽²⁾ 0100b: SPI0_CSOUT⁽²⁾ 0101b: SPI1_CSOUT⁽²⁾ 0110b: RES 0111b: RES 1000b: LIN_Tx⁽³⁾ 1001b: BEMF_COMP_TACHO⁽⁴⁾ 1010b: T1_Trigger⁽⁵⁾ 1011b: T2_Trigger⁽⁵⁾ 1100b: T3_Trigger⁽⁵⁾ 1101b: T4_Trigger⁽⁵⁾ 1110b: PWM_Centre⁽⁶⁾ 1111b: Any Interrupt⁽⁷⁾
DRVS ⁽⁸⁾	8	Output Drive Strength 0: Normal mode (default) 1: Boost mode
GPI_INTCFG[2:0]	11:9	GPI Pin Interrupt Trigger Configuration. (When pin is configured as an input only) 000: Rising edge (default) 001: Falling edge 010: Rising or falling edge 011: Level high 100: Level low 101-111: Rising edge
GPI_INTEN	12	GPI interrupt enable bit (when pin configured as input only) Upon detection of the event selected by GPI_INTCFG[2:0] when pin configured as input, the interrupt will be generated if GPI_INTEN is set to 1. 0: Interrupt disabled (default) 1: Interrupt enabled
AOSEL	13	Application Output Selection Driving pin PDx ⁽⁹⁾ 0: Manual drive. Output on pin PDx follows bit PDOOUT[x] from the GPIO_PDOOUT register (default) 1: output on pin PDx is defined by source, ASRC, selected by ASRCSEL[3:0]
Notes:		(1) ASRCSEL decides the final output on the GPIO pin when FOSEL is set to 0 and AOSEL is set 1. Check output selection for more information. (2) Output from serial communication Interface (UART and SPI) (3) Output from LIN Interface (4) Output from advanced motor control timer unit (AMCTU) (5) Output from general purpose timer unit (GTU) (6) Output from pulse width modulation unit (PWMU) (7) If there is an interrupt generated by any one of the peripherals, this output is set to logic high. When no interrupt exists, this bit is set to logic low. (8) Normal mode driver output is a maximum of 1 mA. Boost mode driver output is up to 4 mA. (9) FOSEL in GPIO_USRCx register must be set to 0 for successful operation of this selection.

GPIO_BLANK0

Reset value = 0x0000

Register Address = 0x40008040

This register defines the blank time period used to avoid miss-detection of short circuit condition. The settings in this register defines the blank time period for GPIO pins PD0 to PD3 when set as an output.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
GPIO3_BT[3:0]				GPIO2_BT[3:0]			
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
GPIO1_BT[3:0]				GPIO0_BT[3:0]			
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
GPIO0_BT[3:0]	3:0	GPIO Blank Time Selects the GPIOx (GPIO0-GPIO3) blanking time. Blanking period is used by the short-circuit protection logic. If the blanking time is set to 0, the short-circuit protection is disabled. 0000: 0 ns /disabled (default) 1000: 1.6 μ s 0001: 200 ns 1001: 1.8 μ s 0010: 400 ns 1010: 2.0 μ s 0011: 600 ns 1011: 2.2 μ s 0100: 800 ns 1100: 2.4 μ s 0101: 1 μ s 1101: 2.6 μ s 0110: 1.2 μ s 1110: 2.8 μ s 0111: 1.4 μ s 1111: 3.0 μ s
GPIO1_BT[3:0]	7:4	
GPIO2_BT[3:0]	11:8	
GPIO3_BT[3:0]	15:12	

GPIO_BLANK1

Reset value = 0x0000

Register Address = 0x40008044

This register defines the blank time period used to avoid miss-detection of short circuit condition. The settings in this register defines the blank time period for GPIO pins PD4 to PD7 when set as an output.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
GPIO3_BT[3:0]				GPIO2_BT[3:0]			
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
GPIO5_BT[3:0]				GPIO4_BT[3:0]			
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
GPIO4_BT[3:0]	3:0	GPIO Blank Time
GPIO5_BT[3:0]	7:4	
GPIO6_BT[3:0]	11:8	
GPIO7_BT[3:0]	15:12	
		Selects the GPIOx (GPIO4-GPIO7) blanking time. Blanking period is used by the short-circuit protection logic. If the blanking time is set to 0, the short-circuit protection is disabled.
		0000: 0 ns / disabled (default) 1000: 1.6 μ s
		0001: 200 ns 1001: 1.8 μ s
		0010: 400 ns 1010: 2.0 μ s
		0011: 600 ns 1011: 2.2 μ s
		0100: 800 ns 1100: 2.4 μ s
		0101: 1 μ s 1101: 2.6 μ s
		0110: 1.2 μ s 1110: 2.8 μ s
		0111: 1.4 μ s 1111: 3.0 μ s

GPIO_PDIN

Reset value = 0x0000

Register Address = 0x40008050

This register shows the current logic level on the GPIO PDx pins.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
GPIL[15:8]							
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
GPIL[7:0]							
R	R	R	R	R	R	R	R

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
GPIL[15:0]	15:0	General Purpose Input Logic Level Logic level on a corresponding GPIO pin, PDx. Bit0 correspond to PD0, bit 1 correspond to PD1 and so on. 0: Logic low 1: Logic high

GPIO_PDOUT

Reset value = 0x0000

Register Address = 0x40008054

This register forces the logic level on the GPIO pins PD0 to PD7 and PD9 to PD15.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
PDSO[15:8]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
PDSO[7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
PDSO[15:0]	7:0	Pin PD Software Output Output logic level on GPIO pin PDx forced by software. Write to this register to manually drive the output on the GPIO pin ^(1,2) . Bit 0 correspond to PD0, bit 1 correspond to PD1,etc. 0: Logic low 1: Logic high
	Notes	(1) To use this register to manually drive the GPIO pins output, AOSEL and FOSEL needs to be set to 0. (2) PD8 is uni-directional input pin only. Writing to bit 8 of PDSO will have no impact on GPIO PD8. (3) The register settings for PD9 to PD15 should be as suggested in output selection section as they are used for die-to-die interface and dedicated peripheral interfaces are created to deal with these die-to-die interface signals. However, if it is desired to force a logic level on these pins, then the settings in this register can be used for pins PD9 to PD15. Care must be taken for pins PD10 to PD15 as they are used for the bridge gate driver interface.

GPIO_STG

Reset value = 0x0000

Register Address = 0x40008058

This register reports the short to ground faults condition.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	Res	Res	Res	Res	Res	Res
R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
STG[7:0]							
R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
STG[7:0]	7:0	Short to Ground Fault Flag This register reports any short-to-GND faults detected when the pin is configured as an output. STG[0] corresponds to PD0, STG[1] corresponds to PD1, etc. 0: No fault detected 1: Pin PDx is shorted to GND

Notes (1) This operation only works for external GPIO pins PD0-PD7. This mode doesn't apply to die-to-die interface pins PD8-PD15.

GPIO_STS

Reset value = 0x0000

Register Address = 0x4000805C

This register reports the short to supply faults condition.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	Res	Res	Res	Res	Res	Res
R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
STS[7:0]							
R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
STS[7:0]	7:0	Short to Supply Fault Flag
		This register reports the any short-to-supply faults detected when the pin is configured as an output. STS[0] corresponds to PD0, STS[1] corresponds to PD1, etc.
		0: No fault detected
		1: Pin PDx is shorted to supply

Notes (1) This operation only works for external GPIO pins PD0-PD7. This mode doesn't apply to die-to-die interface pins PD8-PD15.

GPIO_STG_INTEN

Reset value = 0x0000

Register Address = 0x40008060

This register enables the interrupt generation for the short to ground fault flag reported in GPIO_STG register.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
Res	Res	Res	Res	Res	Res	Res	Res
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
STG_EN[7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
STG_EN[7:0]	7:0	Short to GND Fault Flag Interrupt Enable Interrupt generation enable bit for short-to-GND fault flag. Fault information is reported in GPIO_STG register regardless of the value of this field. Only interrupt generation is affected by changing this field. Bit 0 enables interrupt generation for STG[0], bit 1 enables interrupt generation for STG[1], etc. 0: Interrupt generation due to STG fault disabled 1: Interrupt generation due to STG fault enabled

Notes (1) This operation only works for external GPIO pins PD0-PD7. This mode doesn't apply to die-to-die interface pins PD8-PD15.

GPIO_STS_INTEN

Reset value = 0x0000

Register Address = 0x40008064

This register enables the interrupt generation for the short to supply fault flag reported in GPIO_STS register.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
STS_EN[7:0]							
R/W	R/W	R/W	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
STS_EN	7:0	Interrupt generation enable bit for short-to-supply fault flag. Fault information is reported in GPIO_STS register regardless of the value of this field. Only interrupt generation is affected by changing this field. 0: Interrupt generation due to STS fault disabled 1: Interrupt generation due to STS fault enabled

Notes (1) This operation only works for external GPIO pins PD0-PD7. This mode doesn't apply to die-to-die interface pins PD8-PD15.

GPIO_EVENT_STATUS

Reset value = 0x0000

Register Address = 0x40008068

This register reports if a selected event defined by GPI_INTCFG in GPIO_CONTROL register is detected. The event detection only works when GPIO is set to input.

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
EDS[7:0]							
R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C	R/1C

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
EDS	7:0	Event Detection Status This register reports if an event selected by GPI_INTCFG has been detected on a corresponding pin. EDS[0] corresponds to PD0, EDS[1] corresponds to PD1, etc. If GPI_INTEN in GPIO_CONTROL register is set to 1, an interrupt is also generated for this field. This operation only works whe the pin is set as input. Interrupt status of GPIO pin Dx. Write '1' to clear the interrupt request. 0: No interrupt request triggered 1: Interrupt request triggered

Notes (1) This operation only works for external GPIO pins PD0-PD7. This mode doesn't apply to die-to-die interface pins PD8-PD15.

GPIO_USRCx

Reset value = 0x0000

x represent the GPIO pin number.

ie: GPIO_USRC0 setting is for PD0

Register Address GPIO_USRC0 = 0x40008FC0
 Register Address GPIO_USRC1 = 0x40008FC4
 Register Address GPIO_USRC2 = 0x40008FC8
 Register Address GPIO_USRC3 = 0x40008FCC
 Register Address GPIO_USRC4 = 0x40008FD0
 Register Address GPIO_USRC5 = 0x40008FD4
 Register Address GPIO_USRC6 = 0x40008FD8
 Register Address GPIO_USRC7 = 0x40008FDC
 Register Address GPIO_USRC8 = 0x40008FE0
 Register Address GPIO_USRC9 = 0x40008FE4
 Register Address GPIO_USRC10=0x40008FE8
 Register Address GPIO_USRC11=0x40008FEC
 Register Address GPIO_USRC12=0x40008FF0
 Register Address GPIO_USRC13=0x40008FF4
 Register Address GPIO_USRC14=0x40008FF8
 Register Address GPIO_USRC15=0x40008FFC

When a GPIO is set to output, this registers selects if the final output on a GPIO pin, PDx, should be generated by PDAO or PDEO signal shown in Figure 1-1. This register also defines the the available external sources that defines the signal PDEO. The external sources available in this register are unique for each pin and are not shared across all 15 GPIOs which is opposite to external sources defined in GPIO_CONTROLx register .

bit15	bit14	bit13	bit12	bit11	bit10	bit9	bit8
-	-	-	-	-	-	-	-
U	U	U	U	U	U	U	U

bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
-	-	-	-	ESRCSEL[2:0]		FOSEL	
U	U	U	R/W	R/W	R/W	R/W	R/W

Legend:

R: Read

W: Write

1C: Write 1 to clear. Writing 0 has no effect

U: Unimplemented bit, read as 0

Res: Reserved bit

Field	Bits	Description
FOSEL	0	Final Output Selection 0: Final output on GPIO pin is set to PDAO ¹ signal 1: Final output on GPIO pin is set to PDEO ¹ signal
ESRCSEL[2:0]	3:1	Evaluation Source Selection This selection choose a signal from ESRC array that defines the signal PDEO. 000 _b : ESRC[8*x] 001 _b : ESRC[8*x+1] 010 _b : ESRC[8*x+2] 011 _b : ESRC[8*x+3] 100 _b : ESRC[8*x+4] 101 _b : ESRC[8*x+5] 110 _b : ESRC[8*x+6] 111 _b : ESRC[8*x+7]

Example:

x represent the setting for a corresponding pin PDx. For example, for the pin PD0, x is 0 and the available sources are selected from sources 0 to 7 from ESRC array in Table 4 and are provided below:

000_b: ESRC[0] = PHASEA_DIS
001_b: ESRC[1] = BEMFCOMPC
010_b: ESRC[2] = Reserved
011_b: ESRC[3] = Reserved
100_b: ESRC[4] = Reserved
101_b: ESRC[5] = ADCANT
110_b: ESRC[6] = LC
111_b: ESRC[7] = NT

Notes (1) PDAO signal depends on the setting on GPIO_CONTROLx register. PDEO signal is chosen from ESRC array selected according to ESRCSEL setting. Check output selection [section](#) to learn more about the final output on a GPIO pin.

Table 4: Evaluation Sources (ESRC) which defines the output on a GPIO pin PDx when FOSEL is set to 1

PD0	RES	LC ²	ADCANT ²	RES	RES	RES	BEMFCOMP ¹	PHASEA_DIS ¹
	ESRC[7]	ESRC[6]	ESRC[5]	ESRC[4]	ESRC[3]	ESRC[2]	ESRC[1]	ESRC[0]
PD1	RES	HA ²	ADCBNT ²	RES	RES	RES	COMMUTSEC[0] ¹	PHASEB_DIS ¹
	ESRC[15]	ESRC[14]	ESRC[13]	ESRC[12]	ESRC[11]	ESRC[10]	ESRC[9]	ESRC[8]
PD2	RES	HB ²	ADCCNT ²	RES	RES	DAU_IRQ ⁴	COMMUTSEC[1] ¹	PHASEC_DIS ¹
	ESRC[23]	ESRC[22]	ESRC[21]	ESRC[20]	ESRC[19]	ESRC[18]	ESRC[17]	ESRC[16]
PD3	RES	HC ²	ADCACT ²	RES	CAU_IRQ ⁵	RES	COMMUTSEC[2] ¹	BEMFCOMPA_FILT ¹
	ESRC[31]	ESRC[30]	ESRC[29]	ESRC[28]	ESRC[27]	ESRC[26]	ESRC[25]	ESRC[24]
PD4	RES	T1_source_in ³	ADCBCT ²	V5UV ⁶	RES	RES	RES	BEMFCOMPB_FILT ¹
	ESRC[39]	ESRC[38]	ESRC[37]	ESRC[36]	ESRC[35]	ESRC[34]	ESRC[33]	ESRC[32]
PD5	RES	T2_source_in ³	ADCCCT ²	V3UV ⁶	RES	RES	AMCT_IRQ ⁴	BEMFCOMP ¹
	ESRC[47]	ESRC[46]	ESRC[45]	ESRC[44]	ESRC[43]	ESRC[42]	ESRC[41]	ESRC[40]
PD6	RES	T3_source_in ³	LA ²	V10UV ⁶	RES	RES	RES	BEMFCOMPA ¹
	ESRC[55]	ESRC[54]	ESRC[53]	ESRC[52]	ESRC[51]	ESRC[50]	ESRC[49]	ESRC[48]
PD7	RES	T4_source_in ³	LB ²	PLL_OK ⁵	RES	RES	RES	BEMFCOMPB ¹
	ESRC[63]	ESRC[62]	ESRC[61]	ESRC[60]	ESRC[59]	ESRC[58]	ESRC[57]	ESRC[56]
PD8	RES	RES	RES	RES	RES	RES	RES	RES
	ESRC[71]	ESRC[70]	ESRC[69]	ESRC[68]	ESRC[67]	ESRC[66]	ESRC[65]	ESRC[64]
PD9	RES	RES	RES	RES	RES	RES	RES	RES
	ESRC[79]	ESRC[78]	ESRC[77]	ESRC[76]	ESRC[75]	ESRC[74]	ESRC[73]	ESRC[72]
PD10	RES	RES	RES	RES	RES	RES	RES	HA ²
	ESRC[87]	ESRC[86]	ESRC[85]	ESRC[84]	ESRC[83]	ESRC[82]	ESRC[81]	ESRC[80]
PD11	RES	RES	RES	RES	RES	RES	RES	LA ²
	ESRC[95]	ESRC[94]	ESRC[93]	ESRC[92]	ESRC[91]	ESRC[90]	ESRC[89]	ESRC[88]
PD12	RES	RES	RES	RES	RES	RES	RES	HB ²
	ESRC[103]	ESRC[102]	ESRC[101]	ESRC[100]	ESRC[99]	ESRC[98]	ESRC[97]	ESRC[96]
PD13	RES	RES	RES	RES	RES	RES	RES	LB ²
	ESRC[111]	ESRC[110]	ESRC[109]	ESRC[108]	ESRC[107]	ESRC[106]	ESRC[105]	ESRC[104]
PD14	RES	RES	RES	RES	RES	RES	RES	HC ²
	ESRC[119]	ESRC[118]	ESRC[117]	ESRC[116]	ESRC[115]	ESRC[114]	ESRC[113]	ESRC[112]
PD15	RES	RES	RES	RES	RES	RES	RES	LC ²
	ESRC[127]	ESRC[126]	ESRC[125]	ESRC[124]	ESRC[123]	ESRC[122]	ESRC[121]	ESRC[120]

- Notes:
- (1) Output from advanced motor control timer unit (AMCTU)
 - (2) Output from pulse width modulation unit (PWMU)
 - (3) Output from general purpose timer unit (GTU)
 - (4) Output from data acquisition unit (DAU)
 - (5) Output from current acquisition unit (CAU)
 - (6) Output from system control unit (SCU)

To learn more about the signals above, refer to a corresponding documentation and search for a desired signal name.

Revision Control

Jan 15/2021	Initial Revision
Nov 11/2021	Updated Logo Added revision control

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